

Bayesian Reasoning for LLM Agents under Uncertain and Conflicting Evidence

Milestone 1 Report

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1. Problem Statement

Large Language Model (LLM) agents can reason using information from multiple sources, but their confidence may not change appropriately when some available information is unreliable, irrelevant, or conflicting.

BayesAgent addresses uncertainty in LLM reasoning by representing the reasoning process using latent variables and a verbalized Probabilistic Graphical Model (vPGM). The framework identifies latent reasoning variables, estimates their posterior probabilities, and uses Bayesian inference to obtain the final prediction and confidence.

This work studies a broader evidence setting where an agent receives multiple pieces of evidence with different levels of reliability and consistency. The main problem is to study how the agent should update its belief and confidence when new evidence is introduced, when evidence is irrelevant or misleading, and when different sources provide conflicting information.

This work studies whether extending the Bayesian reasoning framework of BayesAgent can provide more appropriate belief updates and better-calibrated confidence under changing evidence.

Motivation

LLM agents increasingly use information from different sources while solving a problem. Treating every piece of information equally may result in incorrect reasoning or inappropriate confidence. Bayesian reasoning provides a natural way to represent uncertainty and update beliefs when new evidence is received. BayesAgent shows that probabilistic reasoning over latent variables can improve confidence calibration in LLMs.

Significance

The project focuses not only on whether the final answer is correct, but also on whether the agent's confidence responds appropriately to changes in evidence. A useful Bayesian reasoning system should increase confidence when reliable evidence supports a belief, decrease confidence when reliable evidence contradicts it, and avoid making large changes when irrelevant information is introduced.

2. Review of Existing Work

The initial literature study focused on Bayesian reasoning, uncertainty estimation, confidence calibration, and probabilistic reasoning in LLMs. The main works studied so far are summarized below.

Work	Problem / Focus	Main Idea	Relevance
BayesAgent (2026)	Uncertainty in LLM reasoning	Verbalized PGM and Bayesian inference over latent reasoning variables	Main base framework
Chain-of-Thought (2022)	Multi-step LLM reasoning	Step-by-step reasoning	Baseline
Guo et al. (2017)	Neural network calibration	Study and evaluation of confidence calibration	Calibration evaluation
ScienceQA (2022)	Science question answering	Science reasoning benchmark	Main evaluation dataset
Confidence Elicitation (2023)	Confidence estimation in LLMs	Methods for obtaining confidence from LLMs	Baseline and comparison
Bayesian Teaching (2026)	Probabilistic reasoning in LLMs	Bayesian teaching for probabilistic reasoning	Related Bayesian reasoning work
Microsoft Research work	Bayesian view of in-context learning	Probabilistic analysis of LLM in-context learning	Theoretical background

Key Observations from Literature

- LLMs can produce useful confidence estimates, but confidence and correctness do not always agree.
- Calibration is important in addition to prediction accuracy.
- BayesAgent shows that latent-variable probabilistic reasoning can improve uncertainty calibration.
- BayesAgent also studies noisy or mismatched information using A-OKVQA.
- Multiple evidence sources can contain different levels of reliability and consistency.
- Sequential evidence introduces the need to update an existing belief when new information arrives.
- Conflicting evidence provides a controlled setting for studying whether an LLM changes its confidence appropriately.
- Bayesian inference provides a suitable framework for representing and updating uncertain beliefs.

3. Current SOTA Position and Research Direction

BayesAgent is the closest work to the proposed project because it directly applies probabilistic graphical modeling and Bayesian inference to LLM reasoning.

The BayesAgent framework contains three main stages: (1) Graphical Structure Discovery, where the LLM identifies latent variables and probabilistic dependencies; (2) Prompting-Based Inference, where the LLM estimates verbalized posterior

distributions for latent variables; and (3) Prediction under Uncertainty, where the posterior information is used to estimate the probability of the final prediction.

The main inference formulation is:

$$P(Y|X) = \sum_Z P(Y|Z) P(Z|X)$$

where X is the observed input, Z represents latent reasoning variables, and Y is the final prediction. BayesAgent reports improved calibration on benchmarks such as ScienceQA and A-OKVQA.

The proposed project uses BayesAgent as the base framework. The research direction is to investigate a more explicit evidence-integration setting.

Research Question

How can the Bayesian reasoning framework of BayesAgent be extended to model multiple pieces of evidence with different reliability and consistency, and update an LLM agent's beliefs and confidence as new or conflicting evidence is received?

Initial PGM Formulation

The proposed setting considers X as the original question or input, $E_1, \dots, E_m$ as available evidence, $R_1, \dots, R_m$ as reliability or quality of evidence, $Z_1, \dots, Z_n$ as latent reasoning variables, and Y as the final answer.

A conceptual inference formulation is:

$$P(Y|X, E_1, \dots, E_m) = \sum_Z P(Y|Z, E_1, \dots, E_m) P(Z|X, E_1, \dots, E_m)$$

The exact graphical structure and reliability representation will be refined during the experimental stage.

Representation

The PGM represents dependencies between the input, evidence, latent reasoning variables, and final answer. The latent variables represent intermediate reasoning factors used by the LLM. Evidence-related variables are introduced to study how different evidence conditions affect the reasoning and final confidence.

Inference

The project will use Bayesian inference to estimate the final answer distribution from the latent reasoning variables and available evidence. For sequential evidence, the project will study the update $P_t(Y) \rightarrow P_{t+1}(Y|E_{t+1})$, allowing the change in belief after receiving new evidence to be measured.

Learning

The initial stage will reproduce the learning and calibration procedure used by BayesAgent. The proposed extension will then investigate how evidence reliability and consistency can be incorporated into the probabilistic model. The exact parameterization and learning procedure will be finalized after the reproduction and baseline experiments.

4. Progress Completed

Area	Status	Progress
Problem understanding	Completed	Studied uncertainty and evidence integration in LLM reasoning.
Problem formulation	Completed	Defined the problem around multiple, unreliable, irrelevant, misleading, and conflicting evidence.
BayesAgent study	Completed	Studied the vPGM structure, latent variables, Bayesian inference, and calibration framework.
PGM representation	Completed	Identified observed variables, evidence variables, latent variables, and final prediction.
Inference understanding	Completed	Studied Bayesian posterior inference used by BayesAgent.
Literature review	Completed / ongoing refinement	Reviewed Bayesian reasoning, calibration, confidence elicitation, and LLM reasoning works.
Research gap identification	Completed	Identified the need to study evidence reliability, conflict, and sequential belief updates.
Dataset planning	Completed	Selected ScienceQA as the main benchmark and A-OKVQA for noisy evidence evaluation.
Baseline planning	Started	Identified direct prompting, CoT, BayesAgent, and BayesVPGM as initial baselines.
Proposed extension	Started	Initial design for multiple and sequential evidence evaluation has been defined.

At this stage, the problem and research direction have been defined. The BayesAgent framework and its PGM formulation have been studied. The datasets, baselines, and initial evaluation measures have also been identified. Implementation and controlled evidence experiments will be developed in the next stage.

5. Planned Experimental Direction

- Reproduce the main BayesAgent experiments on the selected benchmarks.
- Reproduce the relevant baseline methods before evaluating the proposed extension.
- Compare direct prompting, Chain-of-Thought, BayesAgent, and BayesVPGM.
- Evaluate the systems using both accuracy and calibration measures.
- Construct controlled evidence settings for reliable, irrelevant, misleading, and conflicting evidence.
- Study how the confidence changes after adding each type of evidence.
- Introduce evidence sequentially and measure belief updates after each new evidence item.
- Study whether reliable supporting evidence increases confidence and reliable contradictory evidence decreases confidence.
- Study whether irrelevant evidence causes unnecessary changes in confidence.
- Investigate how evidence reliability and consistency can be represented in the PGM.
- Perform ablation experiments to understand the contribution of Bayesian updating, latent variables, and evidence-related components.
- Perform hyperparameter experiments only after the reproduction and core pipeline are stable.

6. Timeline and Next Milestones

Phase	Planned Work	Status
Phase 1 (Weeks 1-3)	Understand LLM uncertainty, define problem, study BayesAgent and recent literature	About to be Complete
Phase 2 (Weeks 4-6)	Reproduce BayesAgent and implement baselines	In progress
Phase 3 (Weeks 7-9)	Introduce multiple evidence and controlled evidence conditions	Upcoming
Phase 4 (Weeks 10-12)	Sequential evidence and Bayesian belief updates	Planned
Phase 5 (Weeks 13-14)	Ablation, calibration analysis, and final comparison	Planned
Phase 6 (Final)	Final paper preparation and presentation	Planned

Immediate Next Steps

- Complete and verify the BayesAgent reproduction.
- Implement and evaluate the selected baselines.
- Finalize the evidence construction procedure.
- Define reliable, irrelevant, misleading, and conflicting evidence settings.
- Implement the first multiple-evidence version of the framework.
- Study sequential evidence and belief updates.
- Generate calibration and confidence-change plots.
- Perform ablation experiments.
- Compare the proposed extension with BayesAgent and other baselines.

7. Expected Evaluation

The evaluation will check both prediction quality and the behaviour of confidence under changing evidence.

Evaluation Aspect	Purpose	Stage
Accuracy	Measure final prediction correctness	Baseline and final evaluation
Expected Calibration Error	Measure agreement between confidence and correctness	Main evaluation
Confidence Change	Measure how confidence changes after new evidence	Evidence experiments
Evidence Sensitivity	Study response to supporting, irrelevant, misleading, and conflicting evidence	Main experiments
Sequential Belief Update	Measure confidence after each new evidence item	Sequential experiments
Mismatch Detection	Study whether inconsistent evidence can be identified	Evidence analysis
Ablation	Measure contribution of individual components	Final experiments

Accuracy

Accuracy will measure whether the final prediction is correct.

Calibration

Expected Calibration Error (ECE) will measure the difference between predicted confidence and observed accuracy. Lower ECE indicates better calibration.

Confidence Change

The project will measure $\Delta C = C_{\text{after}} - C_{\text{before}}$ to show whether confidence changes after new evidence.

Evidence Sensitivity

The system will be evaluated under different evidence conditions. Supporting evidence should generally increase confidence. Strong contradictory evidence should generally decrease confidence. Irrelevant evidence should not cause a large unnecessary change. Misleading evidence will be studied to determine whether the system can detect or reduce its influence.

Sequential Evidence

Evidence will be provided one piece at a time. The belief and confidence after each step will be recorded to study whether the system performs appropriate Bayesian updates.

8. Conclusion

The first milestone established the research direction for Bayesian reasoning under uncertain and conflicting evidence. The work has moved from problem understanding to a clear research question, literature review, BayesAgent study, PGM formulation, and initial experimental planning.

BayesAgent provides the main technical foundation by representing LLM reasoning using latent variables and applying Bayesian inference for uncertainty estimation. The proposed work extends this direction by studying multiple evidence sources with different reliability and consistency.

The next milestone will focus on reproducing BayesAgent, implementing the selected baselines, and developing controlled experiments for reliable, irrelevant, misleading, and conflicting evidence. This will be followed by sequential belief-update experiments, calibration analysis, and ablation studies.

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